

Detection and Correction of Silent Decision Degradation in AI-Assisted Logistics Systems

Project Description

Modern AI-assisted logistics and supply chain systems are increasingly deployed in high-stakes environments such as defense logistics, transportation planning, and large-scale industrial operations. While significant research focuses on improving model accuracy, routing efficiency, or forecasting performance, a critical and under-explored issue remains: **AI systems can degrade silently over time while continuing to produce seemingly reasonable outputs.**

Unlike explicit failures (system crashes or accuracy collapse), silent decision degradation occurs when an AI system's internal assumptions drift due to partial observability, delayed data, distribution shifts, or changing operational priorities. In such cases, standard evaluation metrics often fail to detect the problem, leading to confident yet systematically incorrect decisions. This phenomenon is especially dangerous in production and defense-oriented systems where delayed or suboptimal decisions accumulate into large operational risks.

This project aims to **identify, quantify, and correct silent decision degradation** in AI-assisted logistics systems through a simulation-driven and prototype-based approach.

Objectives

1. To design a realistic AI-assisted logistics decision system inspired by real-world supply chain and transport operations.
2. To simulate controlled operational disturbances such as data delays, partial information loss, priority inversions, and distribution shifts.
3. To demonstrate that conventional performance metrics fail to reliably detect gradual decision degradation.
4. To develop an AI-based degradation detection layer that monitors structural and behavioral patterns rather than raw accuracy.
5. To implement a correction mechanism that enables the system to self-adapt and recover decision quality over time.

Methodology

The project will be implemented as a multi-layer prototype system:

- **Base Decision System:**
A logistics decision framework (routing, prioritization, or allocation) built using AI/ML

and rule-based components, drawing inspiration from existing defense and supply chain systems.

- **Failure Injection Engine:**

A simulation module that introduces controlled disturbances such as delayed updates, incomplete data, and dynamic priority changes to emulate real operational conditions.

- **Degradation Detection Layer:**

An analytical layer that monitors non-traditional indicators including decision volatility, constraint violation trends, confidence-outcome divergence, and structural changes in system graphs.

- **Correction and Adaptation Layer:**

A hybrid AI mechanism combining statistical thresholds, ensemble disagreement, and rule-based arbitration to recalibrate system behavior and restore decision reliability.

Expected Outcomes

- A working prototype demonstrating silent decision degradation under realistic conditions.
- Quantitative evidence showing the limitations of conventional evaluation metrics.
- A novel detection and correction framework that improves long-term decision reliability.
- Clear experimental results and visualizations supporting system behavior before and after correction.
- A structured report linking Production Engineering principles with AI system design.

Significance

This project shifts focus from optimizing isolated AI models to ensuring **decision integrity in production systems**. The work is directly relevant to defense logistics, large-scale supply chains, and AI deployment in real-world operations, making it valuable for both academic evaluation and industry or government applications.